

Haotian Yao

Shenzhen, China · yaoht100@outlook.com

EDUCATION



AWARDS & CERTIFICATES



RESEARCH & EXTRA-CURRICULAR



Cosmic Ray Detection — Data Analysis & Machine Learning Summer 2024, Shanghai

- Conducted cosmic ray experiment with timestamp, pressure, and coincidence event data collection
- Examined relationship between atmospheric pressure and muon detection rate via linear regression
- Optimized data processing, improved coincidence event identification, and reduced noise

Underwater Robots & Ocean Exploration Project Spring 2024, Shenzhen

- Interdisciplinary learning in mechanical design, data analysis, and electronics
- Built and tested a remotely operated robot for underwater environmental data collection
- Applied control algorithms to strengthen signal stability and improve precision

University of Hong Kong Summer Programme Summer 2025, Hong Kong

- "Data and Systems Engineering in Response to Global Challenges" programme
- Explored 3D printing, robotics, and smart airport systems
- Collected and analyzed robotic motion data with mBlock programming and visualization
- Applied data to ergonomic device optimization; delivered presentations

edX (IBM) MOOC — Machine Learning with Python 2024 – 2025

- Completed assessments and earned certificates in statistical modeling and machine learning
- Applied ML data in healthcare and finance domains, focused on model accuracy

Chinese Academy of Sciences Research Programme Winter 2023, Shenzhen

- Attended lectures on brain-computer interface research with Dr. Han
- Explored zero-gravity backpacks and real-world scientific applications

Shenzhen Senior High School — CS Club & Basketball Club2022 – Present

- CS Club: Organized coding workshops covering Python and algorithmic problem-solving
- Basketball Club: Team training, competitions, and hosting inner-school matches

VOLUNTEER EXPERIENCE^

Mission For Education Equality2023 – Present

Founder

- Online tutoring initiative supporting children in urban villages
- Provided mathematics and English courses

Volunteer Teaching — Zhaluomude School2024, Inner Mongolia

- Taught English and Science classes to local students
- Designed engaging activities to motivate learning

SKILLS^

PythonJupyter NotebookmBlockMS OfficeBambu StudioXRCC

Mandarin (Native)English (Fluent)

PERSONAL STATEMENT^

As an NBA fan and game analysis blogger, I have always sought to compare players' contributions from an objective and quantitative perspective. My journey began with downloading raw data and organizing it in Excel, but I soon realized that changes in rules and tactics made the original data insufficient — I needed a more systematic analytical model.

This curiosity led me through MOOC machine learning projects, cosmic ray detection research, and underwater robotics engineering. In each experience, I strengthened my conviction that data holds discoverable patterns and that statistical modeling and algorithmic optimization are tools not only for processing complexity, but also for revealing underlying physical laws.

My goal is to become a data mining engineer who develops data-driven strategies and solves related global challenges — bridging rigorous mathematical foundations with real-world interdisciplinary applications.

© 2026 Haotian Yao